Anish Bhat

AI Engineer | Generative AI | LLMs | RAG | Agentic AI

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Professional Summary

AI/GenAI Engineer specializing in Large Language Models, Retrieval-Augmented Generation, and Agentic AI systems, with hands-on experience building and deploying production-grade AI applications across cloud and edge environments. Skilled in designing RAG pipelines, fine-tuning transformer-based models, building multi-agent orchestration systems, and developing REST APIs with FastAPI. Experienced across the full ML lifecycle, from data processing and model quantization to backend integration and containerized deployment. Author of 3 peer-reviewed papers on model optimization and multimodal document intelligence.

Technical Skills

- **Languages:** Python, JavaScript, SQL, HTML/CSS
- **AI / Machine Learning:** Machine Learning, Deep Learning, NLP, Computer Vision, Transformers, Fine-tuning, Model Evaluation, Model Quantization
- **Generative AI:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings, Vector Search, AI Agents / Agentic AI, Multi-Agent Systems, Function Calling, Conversational AI, Voice AI
- **Frameworks & Libraries:** PyTorch, TensorFlow, Hugging Face Transformers, LangChain, OpenCV, NumPy, Pandas
- **Backend Development:** FastAPI, Flask, Next.js, REST APIs, JWT Authentication, Webhooks, Async Workflows
- **Databases:** PostgreSQL, MongoDB, Pinecone (Vector DB), Firebase, Supabase
- **Cloud & DevOps:** Google Cloud Platform (GCP), Docker, Kubernetes, Vercel, TensorFlow Lite, Edge AI Deployment, Git, GitHub, CI/CD
- **Automation & AI Platforms:** n8n, Vapi, Twilio, ElevenLabs, WhatsApp Automation
- **Engineering Fundamentals:** System Design, OOP, DBMS, Debugging, Root Cause Analysis, Clean Code Principles

Experience

Embedded AI Intern — Feb 2025 – Apr 2025

Decibels Lab — *Bengaluru, India*

- Trained and evaluated YOLOv11 object detection models on KITTI datasets using statistical validation metrics, achieving a 65% reduction in model size through structured pruning and INT8 quantization.
- Engineered preprocessing and feature-extraction pipelines for autonomous-driving datasets, boosting realtime detection throughput by 20% across Raspberry Pi and Jetson Nano edge devices.
- Structured a scalable backend ML pipeline integrating realtime object detection, lane detection, and sensor fusion at approximately 210ms end-to-end latency, resolving synchronization and inference bottlenecks.
- Containerized embedded AI services with Docker and exposed REST APIs for telemetry and monitoring, maintaining structured GitHub branching and deployment workflows.

AI Engineer & Automation Developer — Dec 2024 – Jun 2026

Independent Freelance — *Remote*

- Built and deployed a full-stack e-commerce platform for a clothing brand using Medusa.js with a customized backend, product catalog, and order-management workflows; owned architecture, integration, and deployment end-to-end. Ran Instagram-led marketing for the brand, growing monthly reach to 200K+ views in 3 months and contributing to an estimated 30% month-over-month revenue increase.
- Developed and deployed a multimodal assistive AI prototype (BLIP-2 vision-language model, STT/TTS pipelines) for realtime scene understanding, integrating stereo cameras and Bluetooth modules; optimized inference for Raspberry Pi 5 edge deployment.
- Architected **Svarra**, a multi-agent AI voice automation platform with 4 specialized agents (FastAPI, Next.js) for inbound/outbound calling, lead qualification, and appointment booking, using event-driven webhook orchestration (n8n, Supabase, Twilio, Vapi, ElevenLabs); built a realtime analytics dashboard for call cost and conversion tracking.
- Delivered AI-assisted digital strategy, ad campaigns, and WhatsApp automation for additional small businesses, improving customer engagement and order handling.

Selected Projects

Agentic RAG Medical Data Extractor — *2025*

- **Problem → Solution:** Manual extraction from large unstructured medical PDFs was slow; created a RAG pipeline with chunking strategies, vector-database indexing, and LLM-driven retrieval for systematized structured summarization.
- **Tech Stack:** Gemini API, LangChain, Pinecone, Docker, Firebase, GCP APIs, n8n. **Impact:** Modular, scalable backend supporting automated ingestion, embedding generation, and semantic retrieval with rate-limiting and context orchestration.

YOLO-OptiMob: Edge AI Optimization Pipeline — Published, SPIN 2025 — *2025*

- **Problem → Solution:** YOLO11 models were too large for mobile deployment; applied 40% L1 unstructured pruning and

INT8 quantization, converting to TensorFlow Lite. **Impact:** Reduced model size from 11.4MB to 2.5MB (78% reduction) while preserving accuracy for Android and low-power edge deployment.

Embedded ADAS Prototype — Published, CIS 2025 *2025*

- **Problem → Solution:** Obstacle-aware navigation on constrained hardware needed low-latency, cloud-free inference; built an embedded ADAS with object detection, lane detection, ultrasonic sensing, and motor control.
- **Tech Stack:** Raspberry Pi, quantized YOLOv12n, Flask, sensor fusion. **Impact:** Achieved 6 FPS realtime inference at 210ms latency; built a Flask telemetry dashboard for live monitoring.

LongDocAI: Quantized Multimodal PDF Summarization & QA — Published, ICCIS 2025 *2025*

- **Problem → Solution:** Long unstructured PDFs needed accurate, mechanized summarization and QA at scale; built a pipeline integrating Donut (OCR-free parsing), BART-large-CNN, and FLAN-T5, fine-tuned on 200k+ paper-summary and 30k+ QA samples.
- **Tech Stack:** Transformers, Hugging Face, PyTorch, LLM.int8 quantization. **Impact:** 15% improvement in ROUGE-L, 28% in METEOR, 85.3% QA accuracy, 70% model-size reduction, 40% lower inference latency.

Education

B.E. Computer Science (Artificial Intelligence), KLE Technological University 2022 – 2026
Hubballi, Karnataka

Certifications & Honors

Certifications: NVIDIA – Getting Started with AI on Jetson Nano (2026) | Neo4j Certified Professional: Graph Data Science (2026) | Neo4j: Building Knowledge Graphs with LLMs (2026)

Honors: 1st Place, College Mechatronics Competition, 130+ teams (semi-autonomous delivery robot) | Hackathon Finalist & Team Lead, HackKarnataka | Winner, Smart India Hackathon (Internal Round)